Lesson 10.2 - How to Identify Type I and Type II Errors

> **Video:** How To Identify Type I and Type II Errors In Statistics by The Organic Chemistry Tutor
> **Chapter:** 10 - Hypothesis Testing - 1 Sample

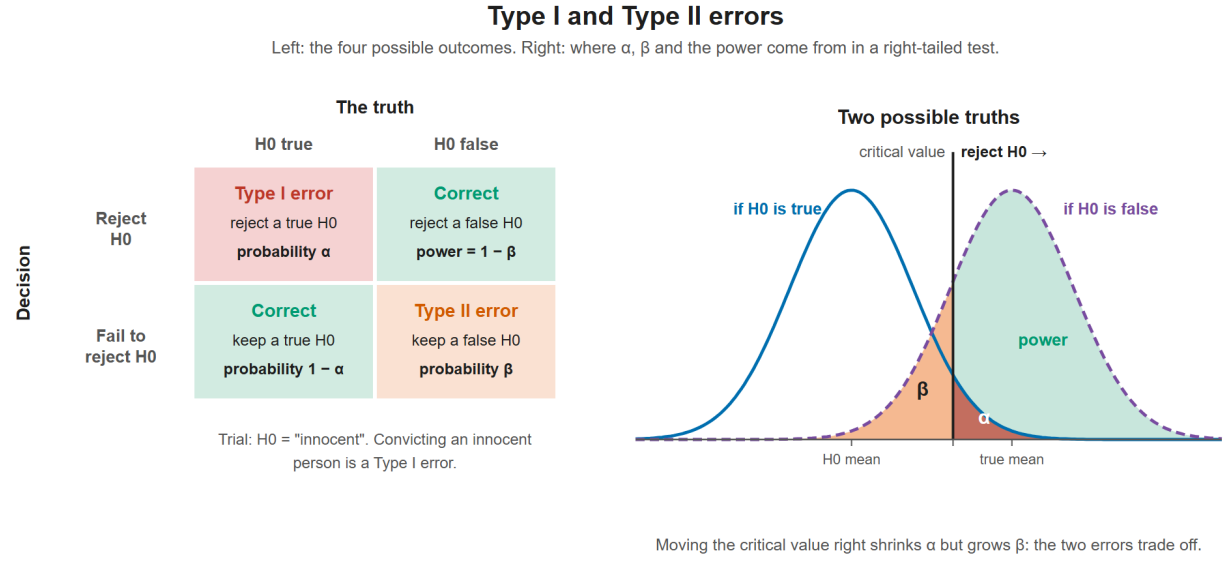
Definitions

Error	What happens	Probability
Type I	**Reject** H_0 when H_0 is actually **true**	α (the significance level)
Type II	**Fail to reject** H_0 when H_0 is actually **false**	β
(correct) Power	Reject H_0 when it is false	$1-\beta$, the **power of the test**

Decision table

	H_0 **is true**	H_0 **is false**
Reject H_0	❌ **Type I error** (α)	✅ Correct (power $= 1-\beta$)
Fail to reject H_0	✅ Correct	❌ **Type II error** (β)

We can't control whether H_0 is true. We only control the decision. (The video sometimes says "accept H_0"; the standard wording is "fail to reject H_0".)



Left: the four combinations of decision and truth. Right: if H0 is true, the red tail beyond the critical value (Î±) is the chance of a false alarm. If H0 is false, the orange area left of the critical value (Î²) is the chance of missing the effect, and the green area is the power.

How to identify the error type

1. Mark each scenario as a **good** decision or a **bad** decision (an error).
2. For each error, ask whether the facts make H_0 **true or false**:
    - H_0 true but rejected → **Type I**
    - H_0 false but not rejected → **Type II**

Example 1 - is John's car safe?

H_0: *John's used car is safe to drive.*

Choice	Situation	Good or bad?	Error
A	Thinks the car is safe, but it is **not** safe	Bad	H_0 false, not rejected → **Type II**
B	Thinks the car is safe, and it **is** safe	Good	—
C	Thinks it may not be safe, and it is **not** safe	Good	—
D	Thinks it may not be safe, but it **is** safe	Bad	H_0 true, rejected → **Type I**

Which error is worse? The Type II error (A): he drives an unsafe car and could be badly hurt. The Type I error (D) only means he stays home unnecessarily.

Example 2 - a criminal trial

H_0: *The defendant is innocent* (presumed innocent).

Choice	Situation	Good or bad?	Error
A	Jury says **guilty**, but he is **innocent**	Bad	H_0 true, rejected → **Type I**
B	Jury says guilty, and he is **not** innocent	Good	—
C	Jury says **not guilty**, but he is **not** innocent	Bad	H_0 false, not rejected → **Type II**
D	Jury says not guilty, and he **is** innocent	Good	—

Which error is worse? Usually the Type I error: an innocent person goes to prison. Letting a guilty person go free (Type II) is also bad, but courts are designed to avoid convicting the innocent. This is partly a judgment call.

Key takeaways

- Type I = **false alarm**: rejecting a true H_0, probability α.
- Type II = **missed detection**: failing to reject a false H_0, probability β.
- Power = $1 - \beta$, the probability of correctly rejecting a false H_0.
- Which error is "worse" depends on the real-world consequences in context.